

genaut — frontier report

source CSV : genaut/logs/2statelaplacc/default.csv
corpus rows: 929

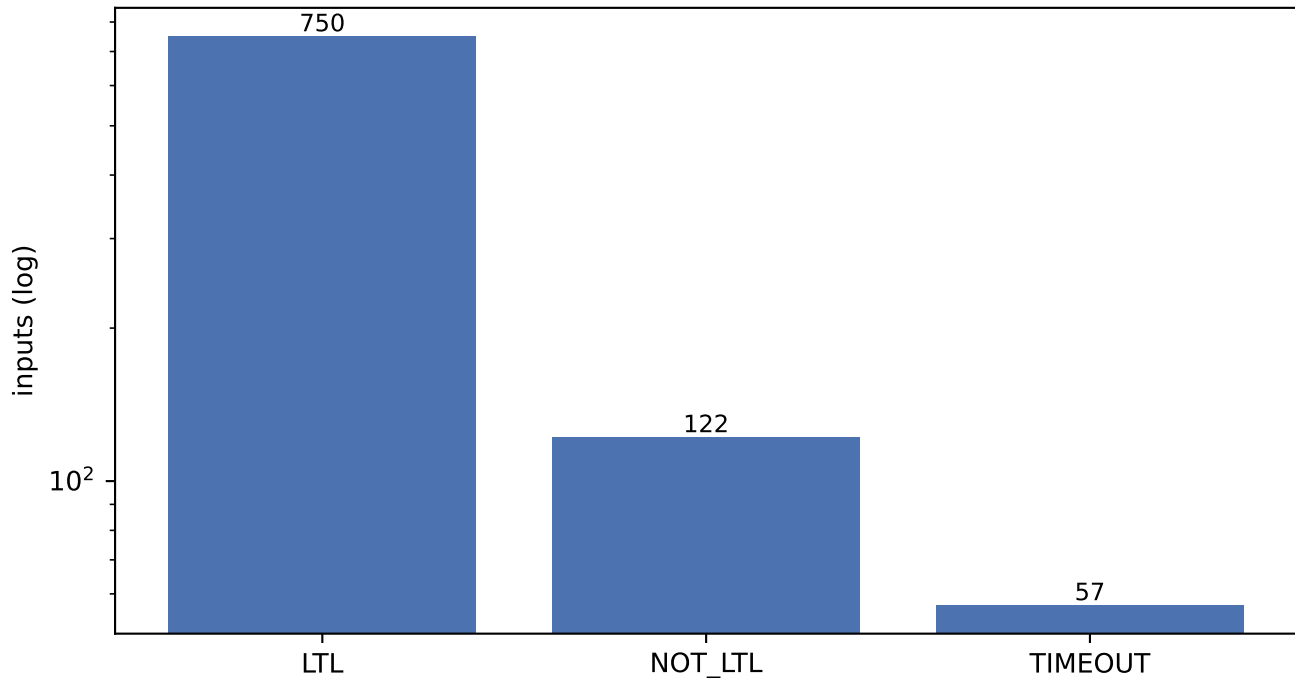
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 929
  LTL-definable (formula built) :   750  ( 80.7%)
  not-LTL (decided)             :   122  ( 13.1%)
  ambiguous (timeout, unknown)  :    57  (  6.1%)
decided (LTL or not-LTL): 872  ( 93.9%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 750 (2 with a gated/unprintable string)

distinct structures (md5): 74

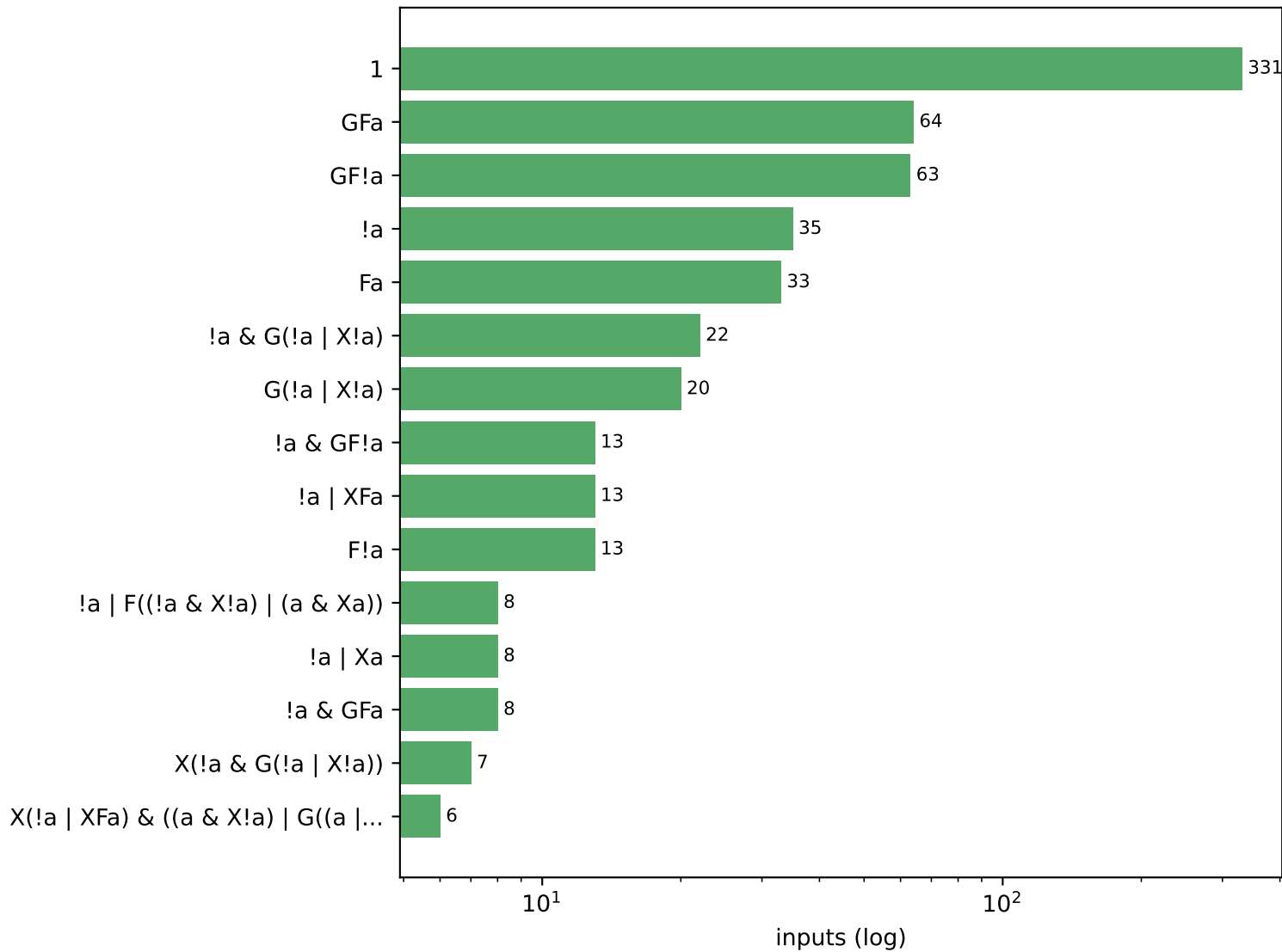
80% of answers covered by the top 10 structure(s)

90% of answers covered by the top 22 structure(s)

top 15 structures:

331	(44.1%)	1
64	(8.5%)	GFa
63	(8.4%)	GF!a
35	(4.7%)	!a
33	(4.4%)	Fa
22	(2.9%)	!a & G(!a X!a)
20	(2.7%)	G(!a X!a)
13	(1.7%)	!a & GF!a
13	(1.7%)	!a XFa
13	(1.7%)	F!a
8	(1.1%)	!a F(!a & X!a) (a & Xa)
8	(1.1%)	!a Xa
8	(1.1%)	!a & GFa
7	(0.9%)	X(!a & G(!a X!a))
6	(0.8%)	X(!a XFa) & ((a & X!a) G((a X((a X(!a...

Top 15 reconstructed structures



Size frontier — flattened formula size

LTL answers with a size: 750

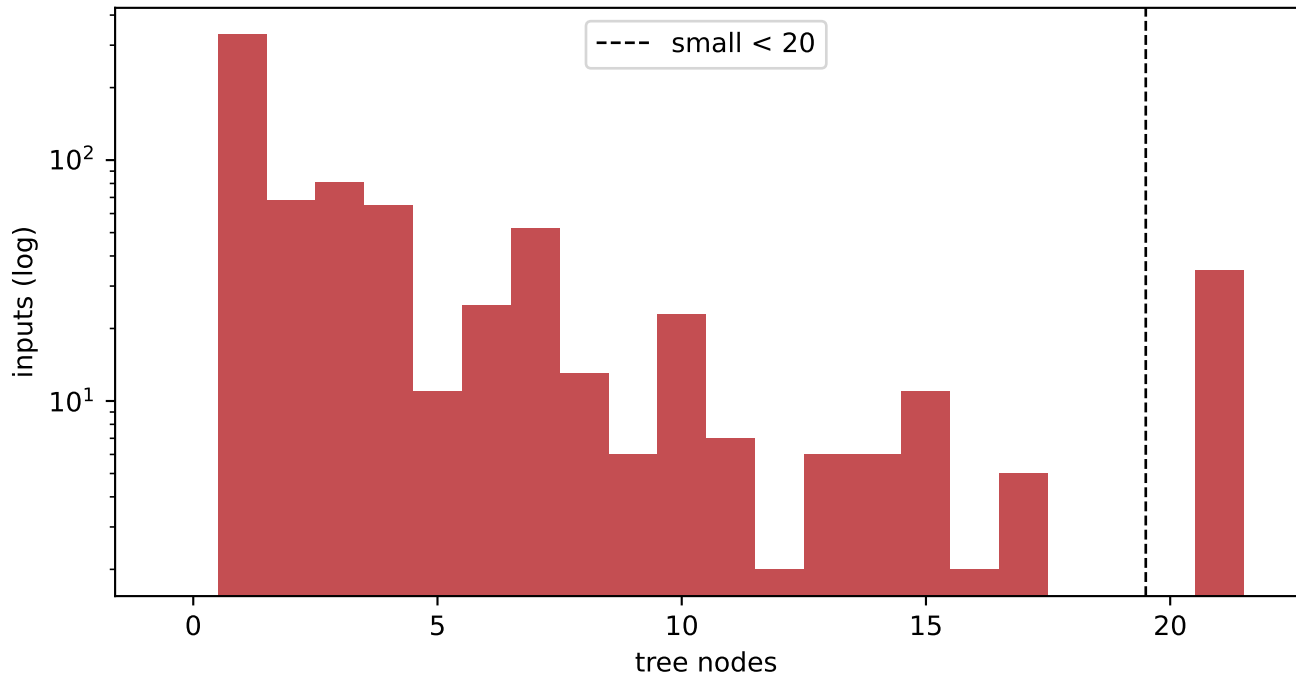
trivial (tree≤1, i.e. true/false): 332 (44.3%)

small (tree<20): 715 (95.3%)

larger (tree≥20): 35 (4.7%)

tree nodes: median 2 mean 777080882.0 max 410075638769

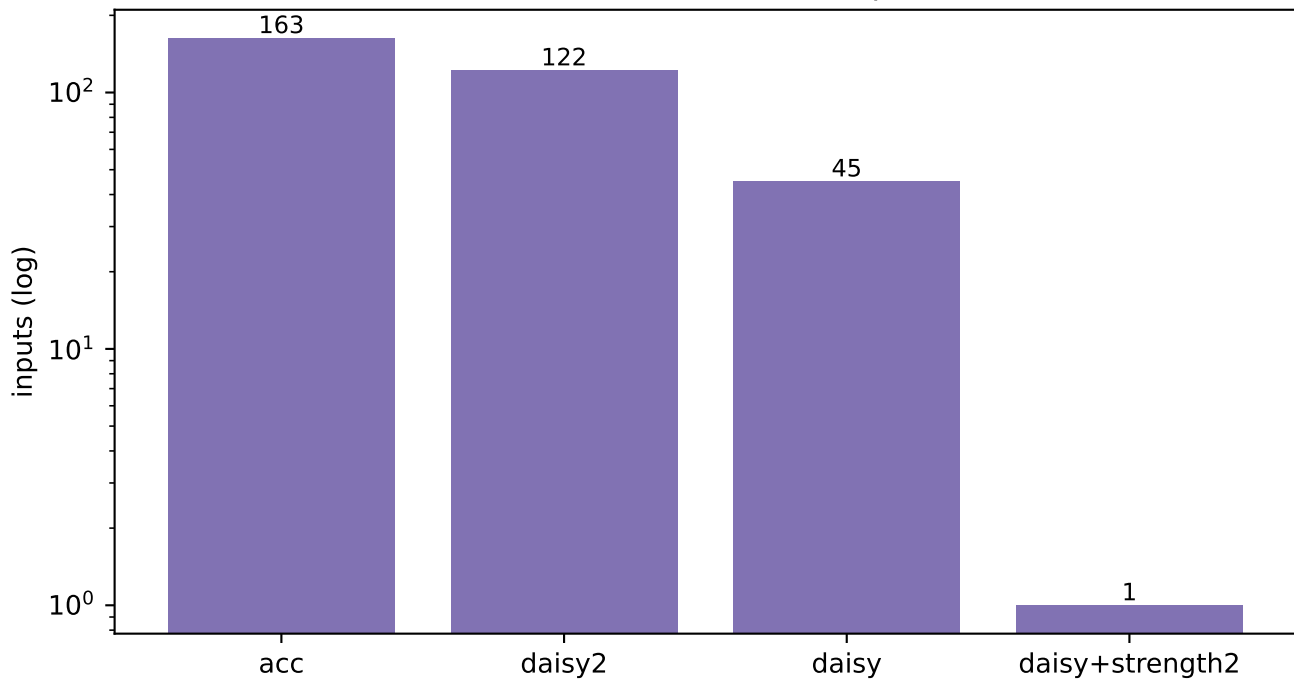
Flattened size (tree nodes); clipped at 21 (35 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 331
  163 ( 49.2%) acc
  122 ( 36.9%) daisy2
   45 ( 13.6%) daisy
    1 (  0.3%) daisy+strength2
build_s over true rows: total 196.8s  mean 0.595s  max 1.400s
```


Routes to `true` (technique)



Build time & timeouts

total build: 1558.4s mean 1.678s max 16.036s

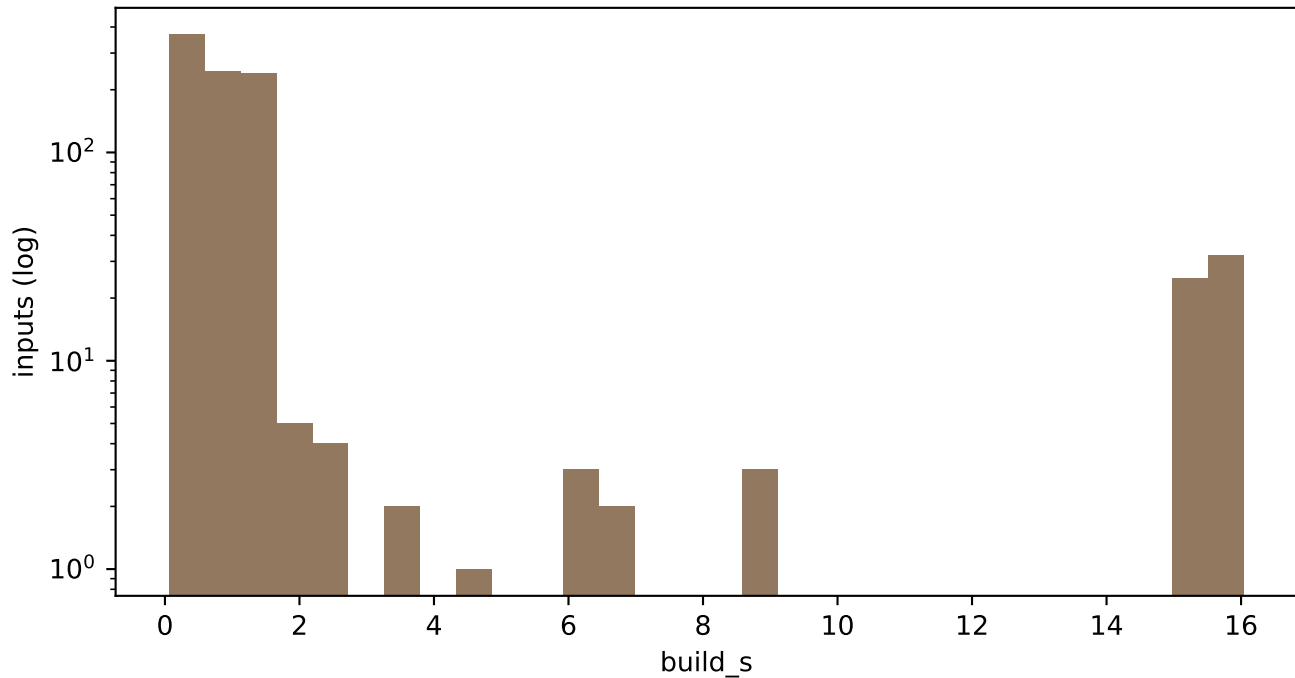
 LTL n= 750 mean 0.710s max 8.810s

 NOT_LTL n= 122 mean 1.125s max 1.189s

timeouts: 57

2statelaplace_00284.hoa
2statelaplace_00300.hoa
2statelaplace_00396.hoa
2statelaplace_01153.hoa
2statelaplace_01154.hoa
2statelaplace_01666.hoa
2statelaplace_03137.hoa
2statelaplace_04370.hoa
2statelaplace_04376.hoa
2statelaplace_04378.hoa
2statelaplace_04424.hoa
2statelaplace_04642.hoa
2statelaplace_04648.hoa
2statelaplace_04650.hoa
2statelaplace_04744.hoa
2statelaplace_05186.hoa
2statelaplace_05248.hoa
2statelaplace_05249.hoa
2statelaplace_05768.hoa
2statelaplace_06208.hoa

Build-time distribution



Verification — spot-oracle validation

validation over 750 LTL answers:
TRUE 748 (99.7%)
SIZE 2 (0.3%)
NOT-equivalent (FALSE): 0 CLEAN